A Three-Stage Hierarchical Soft-Fusion Framework for Cryptocurrency Trading Signal Classification

BBL514E Pattern Recognition — Term Project

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Problem & Contributions

Why is direct Buy/Hold/Sell classification hard?

- Crypto is non-stationary; regimes shift on macro shocks.
- Flat classifiers learn idiosyncratic relationships that fail across regimes.

Our approach. Decompose the decision hierarchically:

1. **Stage 1** — local trend (ZigZag-labelled)
2. **Stage 2** — macro regime (rule-based FSM)
3. **Stage 3** — signal head that soft-fuses the upstream posteriors

Three contributions:

- A fully ablated 3-stage soft-fusion pipeline (BTC + ETH).
- An asset-specific architecture finding (no-free-lunch).
- Hold-aware trading rules turn F1 $= 0.37$ into Sharpe $= 1.15$.

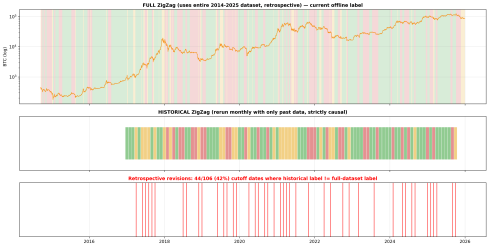
Dataset & Stage 1 Design

Data

- BTC 2014–2025 (4,094 d), ETH 2017–2025 (3,060 d)
- 22 macro covariates (Yahoo + FRED) with publication-release lags
- **Zero look-ahead leakage**, asserted in code

Stage 1 label — ZigZag, 10% dev., $\geq$ 15-d seg.

- Classes: *uptrend* / *range* / *downtrend*
- 14 causal features *(RSI, MACD, ADX, %B, ATR, OBV, log-returns)*



ZigZag label is revisable.

106 causal cutoffs vs. offline label: **42%** disagree

$\Rightarrow$ ML classifier needed.

Stage 1 Results — Frame F1 understates segment quality

Optuna-tuned best per asset

- Random Forest wins both
- BTC: $F1_m$ **0.563**, AUC **0.76**
- ETH: $F1_m$ **0.571**, AUC **0.75**

All 8 tuned configs cleared F1 $\geq$ 0.50 *gate.*

Segment-level finding (20-d rolling mode):

- Majority-vote consistency = **0.77** BTC, **0.83** ETH
- In-segment dominant-class match $>$ **80%**

Stage-1 BTC ground-truth vs. predicted (Random Forest, tuned).

Stage 2 — Composite Macro Finite-State Machine

Why a rule-based FSM?

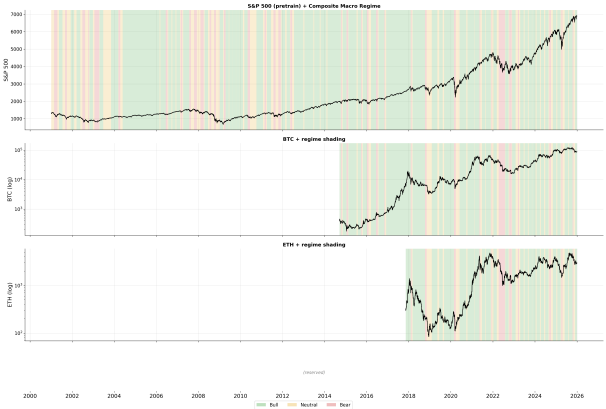
Unsupervised approaches first tried and rejected:

- K-Means: failed on 2008 GFC
- Constrained K-Means: noisy
- HMM: unstable transitions
- GMM: 2024–25 stickiness

FSM design

- 8 rules + 1 re-entry guard
- 5 macro features *(VIX z, yield-curve, ΔVIX, DXY z, M2 YoY)*
- Output: $\hat{p}_{\text{regime}} + \tau_t$

Fires on '08 GFC, '20 COVID, '22 bear, '25



Regime timeline (Bear / Neutral / Bull) over S&P 500, BTC, ETH.

Stage 3 — Soft-Fusion Architecture

Soft fusion — pass the full posterior vectors, not hard labels:

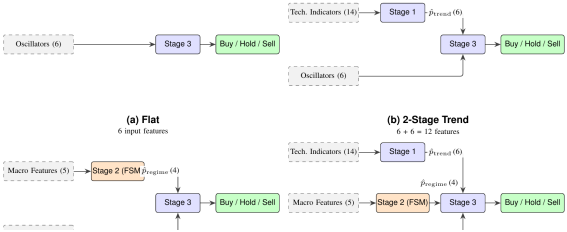
$$z_t = \left[\, \hat{p}_{\text{trend}}(t),\ \hat{p}^{(\text{smooth-10})}_{\text{trend}}(t),\ \hat{p}_{\text{regime}}(t),\ \tau_t,\ o_t \,\right] \in \mathbb{R}^{16}$$

o_t: 6 oscillators (RSI, MACD, %B, ATR, CCI, WilliamsR).

Label (causal):

$$y_t = \begin{cases} \text{Buy} & \text{if } \frac{P_{t+5}-P_t}{P_t} > k\hat{\sigma}_t \\ \text{Sell} & \text{if } \frac{P_{t+5}-P_t}{P_t} < -k\hat{\sigma}_t \\ \text{Hold} & \text{otherwise} \end{cases}$$

$\hat{\sigma}_t$: 20-d past-only rolling σ; $k = 0.5$.



Four architectures ablated: Flat → 2-Stage Trend → 2-Stage Macro → 3-Stage Full.

Experimental Setup — Walk-Forward + Optuna

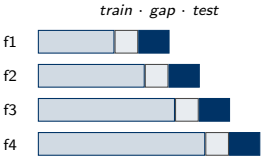
Outer CV — expanding-window walk-forward

- Val window $v = 200\,\mathrm{d}$; advance by v (non-overlap).
- Gap $g = 10\,\mathrm{d} > $ label horizon $h = 5\,\mathrm{d} \Rightarrow$ no leakage.
- Folds: BTC 16/12 (S1/S3); ETH 10/6.

Inner CV — 5-fold walk-forward for Optuna

- Folds span regimes: 2017, 2019, 2021, 2023, 2025.
- TPE + MedianPruner, 30 trials per model/stage.

Backtest — 0.1% one-way cost; 3 rules: *stateful · defensive reset · prob.-weighted.*



Expanding-window: $g = 10\,\mathrm{d}$, $v = 200\,\mathrm{d}$.

1 h 43 m full ablation (M3 MacBook Air)

4 archs × 4 models × 2 assets × 30 trials.

Architecture Ablation — The Asset-Specific Optimum

Asset	Architecture *(best ⟨rule, model⟩)*	Sharpe	Return	MaxDD	$F1_m$
BTC	Flat *(state., xgb)*	0.93	+1565%	−75%	0.349
BTC	2-Stage Trend *(probw., lgbm)*	1.08	+539%	−28%	0.347
BTC	2-Stage Macro *(state., rf)*	0.98	+1606%	−53%	0.356
BTC	**3-Stage Full *(state., xgb)***	**1.15**	**+2901%**	**−46%**	**0.367**
BTC	Buy & Hold	0.95	+2972%	−77%	—
ETH	**Flat *(probw., lgbm)***	**0.52**	**+26%**	**−18%**	0.347
ETH	2-Stage Trend *(defens., xgb)*	0.39	+43%	−55%	0.361
ETH	2-Stage Macro *(state., mlp)*	0.47	+69%	−51%	0.343
ETH	3-Stage Full *(probw., lgbm)*	0.34	+19%	−18%	0.350
ETH	Buy & Hold	0.26	−7%	−72%	—

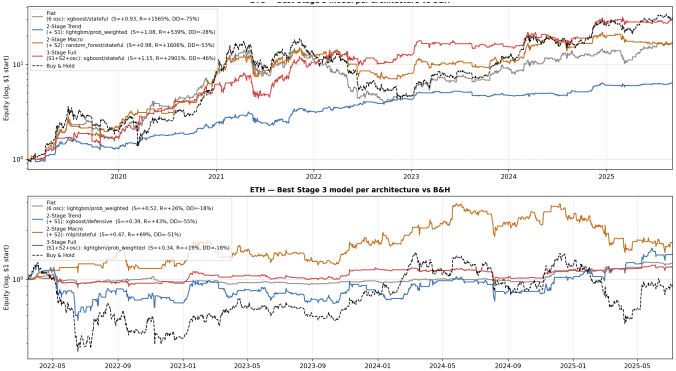
BTC: every hierarchical variant beats Flat ⇒ hierarchy adds value.

ETH: Flat beats 3-Stage ⇒ hierarchy overfits the shorter 5-yr history.

No-free-lunch

Depth is asset-specific. Validate per asset and per dataset size.

Equity Curves — The Money Slide



Best architecture per ablation vs. Buy & Hold. BTC (top), ETH (bottom).

BTC — 3-Stage Full pulls ahead in the **2022 bear** via Hold-class abstention. Drawdown: 77% → **46%**.

ETH — B&H loses 7%; every architecture beats it. Hold class as a **risk filter** ⇒ F1 = 0.37 yields Sharpe = 1.15.

Key Takeaways

1. Frame $\neq$ Trade

Macro-F1 0.37, AUC 0.53 $\Longrightarrow$ **Sharpe** 1.15 via Hold-aware rules.

2. Architecture is asset-specific

3-Stage wins BTC; Flat wins ETH. No universal hierarchy in financial ML.

3. Reproducible

Walk-forward CV + Optuna + Docker. Code, OOF predictions, FastAPI demo all public.

$\Rightarrow$ **Now the live demo.**

Live Demo

FastAPI back-end ▪ Chart.js front-end ▪ one Docker container

`http://localhost:8000`

Stage 1 posterior $\rightarrow$ *Stage 2 regime* $\rightarrow$ *Stage 3 signal* $\rightarrow$ *equity curve*